Fiscal Year:	FY 2011	**Task Last Updated:**	FY 07/27/2011
PI Name:	T'ien, James S. Ph.D.		
Project Title:	Research Opportunities in Combustion Science		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jst2@case.edu	**Fax:**	FY 216 368-3007
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City:	Cleveland	**State:**	OH
Zip Code:	44106-1712	**Congressional District:**	11
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Combustion Science NNH09ZTT001N
Start Date:	02/01/2010	**End Date:**	01/31/2015
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Olson, Sandra	**Contact Phone:**	216-433-2859
Contact Email:	Sandra.Olson@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date is now 1/31/2015 per NSSC information (Ed., 3/12/2013)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Ferkul, Paul (Universities Space Research Association) Olson, Sandra (NASA Glenn Research Center)		
Grant/Contract No.:	NNX10AD99A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	A ground-based experimental and theoretical project for flame spread and extinction over solid materials is proposed with the intention of leading to a flight experiment in the space station. The research will be fundamental in nature but has practical applications to spacecraft fire safety. The objectives are (1) to determine the flame spread and growth characteristics of solids in concurrent flow in reduced gravity, (2) to determine the mechanisms of flame decay and extinction, (3) to establish a high-fidelity numerical model that can simulate both normal and reduced gravity flames, including the interpretation of the results of NASA-STD-6001A test 1. We intend to study both fabric and thick solid samples. This ground-based research is to extend the previous experimental works on solid flame spread in reduced gravity that were almost exclusively limited to very thin samples and to enhance the model capability to a three-dimensional transient formulation. The ground-based experiments will utilize both the normal-gravity and zero-gravity facilities at NASA Glenn Research center. Considerable effort will be spent to plan a space experiment that has to be carried out in the currently available facility in the International Space Station (i.e. CIR). In addition to James T'ien (PI), Dr. Paul Ferkul (NCSER) and Dr. Sandra Olson (NASA GRC) will participate as Co-Is in the project.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This is a fundamental study on the flammability of solids. The understanding to be achieved will help us to deal with fire safety on earth and in space.
Task Progress:	Ignition of a PMMA sphere using a hot wire igniter in normal gravity followed by transition into a forced convective microgravity environment is studied experimentally in NASA's 5.18 second drop tower and modeled using a transient numerical code. The flame is ignited at the forward stagnation region, and then the igniter is retracted. The flame is allowed to spread around the PMMA sphere before release into microgravity. The flame transitions to the new environment quickly, although the solid phase response time is longer than the available drop time. Depending on the oxygen concentration, the forced flow velocity and the degree of solid interior heating, the transition flame is either stabilized around the sphere or shrinks toward the forward stagnation point in a decaying mode. The transient numerical model reproduces the experimental sequences and provides more insight as microgravity computation can be performed in longer time durations.
Bibliography Type:	Description: (Last Updated: 09/19/2025)
Papers from Meeting Proceedings	Olson SL, Johnston M, Hsu SY, T'ien JS, Ferkul PV. "Experiments and Modeling of Normal Gravity Ignition and Transition to Microgravity For A PMMA Sphere In A Forced Convective Flow." 7th US National Technical Meeting of the Combustion Institute, Atlanta, GA, March 20-23, 2011. 2011 US National Combustion Institute Papers. Red Hook, NY : Curran Associates, Inc., 2011. p. 1729-1734. , Jun-2011